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A New Unit Root Test with Two Structural Breaks in Level and Slope at Unknown Time

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A New Unit Root Test with Two Structural Breaks in Level and Slope at Unknown Time

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Abstract

In this paper we propose a new ADF-type test for unit roots which accounts for two structural breaks. We consider two different specifications: (a) two breaks in the level of a trending series; and (b) two breaks in the level and slope of trending data. The breaks whose time of occurrence is assumed to be unknown are modelled as innovational outliers and thus take effect gradually. Using Monte Carlo simulations, we show that our proposed test has correct size, stable power, and identifies the structural breaks accurately.

1 Introduction

The unit root hypothesis has both theoretical and empirical implications for economic theory and modelling. This is one reason for the popularity of unit root tests and a key motivation for methodological innovations. Perron (1989) showed that ignoring a structural break, as is the case with the Dickey and Fuller (DF), can lead to the false acceptance of the unit root null hypothesis. The effect of structural breaks on the performance of the DF unit root test is discussed intensively in the literature. This branch of the literature emphasizes the power reductions of the DF-test if a break occurs under the alternative hypothesis (see, for instance, Perron, 1989; and Rappoport and Reichlin, 1989). In order to handle this problem, Perron (1989) augments the ADF test regression with dummy variables accounting for the break.

In this paper our goal is to extend the literature on unit root tests with structural breaks. Extensions of Perron (1989) have been made by Zivot and Andrews (ZA, 1992) and Perron (1997), *inter alia*, through accounting for an endogenous structural break, and by Lumsdaine and Papell (LP, 1997) through accounting for two structural breaks. However, Lee and Strazicich (LS, 2001, 2003) show that these ADF-type unit root tests which either do not allow for a break under the null as ZA and LP or model the break as an innovational outlier (IO) as Perron (1997) suffer from severe spurious

rejections in finite samples when a break is present under the null hypothesis. Because the spurious rejections are not present in the case of a known break point, LS (2001) identify the inaccurate estimation of the break date as source of the spurious rejections. Judging it difficult to find a convenient remedy to the problem of spurious rejections for ADF-type unit root tests, LS (2003, 2004) follow a different route by proposing a minimum Lagrange Multiplier (LM) unit root test which do not suffer from spurious rejections of this kind. Though, Popp (2008) has pointed out that these spurious rejections are not a general feature of ADF-type unit root tests. Rather, the root of the problem of spurious rejections is that the parameters of the test regression have different interpretations under the null and alternative hypothesis, cf. Schmidt and Phillips (SP, 1992), which is crucial since the parameters have implications for the selection of the structural break date. Following SP (1992), this can be avoided by formulating the data generating process (DGP) as an unobserved components model which allows us to generate a new ADF-type unit root test for the case of IOs. An interesting feature of the new test is that the critical values of the test assuming unknown break dates converges with increasing sample size to the critical values when the break points are known.

We organise the balance of the paper as follows. In section 2, we discuss our proposed new test. In section 3, we assess the size and power properties

of our test. Because the spurious rejections are a feature especially in finite samples, we show the favorable properties of the new test by Monte Carlo simulations. In section 4, we demonstrate the applicability of our new test using the Nelson and Plosser dataset and an up-dated post-war dataset that includes 32 macroeconomic data series for the USA. In section 5, we provide some concluding remarks.

2 Models and test statistics

Following SP (1992), we consider an unobserved components model to represent the DGP. The DGP of a time series y_t has two components, a deterministic component (d_t) and a stochastic component (u_t), as follows:

$$y_t = d_t + u_t, \tag{1}$$

$$u_t = \rho u_{t-1} + \varepsilon_t, \tag{2}$$

$$\varepsilon_t = \Psi^*(L)e_t = A^*(L)^{-1}B(L)e_t, \tag{3}$$

with $e_t \sim iid(0, \sigma_e^2)$. It is assumed that the roots of the lag polynomials $A^*(L)$ and $B(L)$ which are of order p and q , respectively, lie outside the unit circle.

We consider two different specifications both for trending data: one allows

for two breaks in level (denoted model 1 or M1) and the other allows for two breaks in level as well as slope (denoted model 2 or M2). Both model specifications differ in how the deterministic component d_t is defined:

$$d_t^{\text{M1}} = \alpha + \beta t + \Psi^*(L) (\theta_1 DU'_{1,t} + \theta_2 DU'_{2,t}), \quad (4)$$

$$d_t^{\text{M2}} = \alpha + \beta t + \Psi^*(L) (\theta_1 DU'_{1,t} + \theta_2 DU'_{2,t} + \gamma_1 DT'_{1,t} + \gamma_2 DT'_{2,t}), \quad (5)$$

with

$$DU'_{i,t} = 1(t > T'_{B,i}), \quad DT'_{i,t} = 1(t > T'_{B,i})(t - T'_{B,i}), \quad i = 1, 2. \quad (6)$$

Here, $T'_{B,i}$, $i = 1, 2$, denote the true break dates. The parameters θ_i and γ_i indicate the magnitude of the level and slope breaks, respectively. The inclusion of $\Psi^*(L)$ in (4) and (5) enables the breaks to occur slowly over time. Specifically, it is assumed that the series responds to shocks to the trend function the way it reacts to shocks to the innovation process e_t , see Vogelsang and Perron (VP, 1998). This approach is called the IO model.

The IO-type test regressions for M1 and M2 to test the unit root null hypothesis can be derived by merging the structural model (1)-(5). The test

equation for M1 has the following form:

$$y_t^{\text{M1}} = \rho y_{t-1} + \alpha_1 + \beta^* t + \theta_1 D(T'_B)_{1,t} + \theta_2 D(T'_B)_{2,t} + \delta_1 DU'_{1,t-1} + \delta_2 DU'_{2,t-1} + \sum_{j=1}^k \beta_j \Delta y_{t-j} + e_t, \quad (7)$$

with $\alpha_1 = \Psi^*(1)^{-1} [(1 - \rho)\alpha + \rho\beta] + \Psi^{*'}(1)^{-1}(1 - \rho)\beta$, $\Psi^{*'}(1)^{-1}$ being the mean lag, $\beta^* = \Psi^*(1)^{-1}(1 - \rho)\beta$, $\phi = \rho - 1$, $\delta_i = -\phi\theta_i$ and $D(T'_B)_{i,t} = 1(t = T'_{B,i} + 1)$, $i = 1, 2$.

The IO-type test regression for M2 is as follows:

$$y_t^{\text{M2}} = \rho y_{t-1} + \alpha^* + \beta^* t + \kappa_1 D(T'_B)_{1,t} + \kappa_2 D(T'_B)_{2,t} + \delta_1^* DU'_{1,t-1} + \delta_2^* DU'_{2,t-1} + \gamma_1^* DT'_{1,t-1} + \gamma_2^* DT'_{2,t-1} + \sum_{j=1}^k \beta_j \Delta y_{t-j} + e_t, \quad (8)$$

where $\kappa_i = (\theta_i + \gamma_i)$, $\delta_i^* = (\gamma_i - \phi\theta_i)$ and $\gamma_i^* = -\phi\gamma_i$, $i = 1, 2$.

In order to test the unit root null hypothesis of $\rho = 1$ against the alternative hypothesis of $\rho < 1$, we use the t-statistics of $\hat{\rho}$, denoted $t_{\hat{\rho}}$, in (7) and (8).

It is worth noting that in contrast to the well-known Perron-type test regressions for the one break case, see e.g. equations (5.1) and (5.2) in VP (1998), the dummy variables $DU'_{i,t}$ and $DT'_{i,t}$ are lagged in (7) and (8). However, for given break dates, both the Perron-type test regressions (augmented

to two breaks) and the test regressions formulated in (7) and (8) produce identical t-values $t_{\hat{\rho}}$.¹ Despite this fact, we favor the use of (7) and (8) because the coefficients of the impulse dummy variable $D(T'_B)_{i,t}$, θ_i for M1 and κ_i for M2, solely comprise the break parameters θ_i and γ_i . This is essential in the situation of an unknown break date in which we want to identify the timing of the break on the basis of estimates of the break parameters.

Because we assume that the true break dates are unknown, $T'_{B,i}$ in equations (7) and (8) has to be substituted by their estimates $\hat{T}_{B,i}$, $i = 1, 2$, in order to conduct the unit root test. The break dates can be selected simultaneously following a grid search procedure. Therefor, we conduct the test regressions for every potential break point combination $(T_{B,1}, T_{B,2})$ and choose that points in time as break dates for which the joint significance of the impulse dummy variable coefficients is maximised, i.e.

$$\left(\hat{T}_{B,1}, \hat{T}_{B,2}\right) = \begin{cases} \arg \max F_{\hat{\theta}_1, \hat{\theta}_2} & , \text{ for model 1} \\ \arg \max F_{\hat{\kappa}_1, \hat{\kappa}_2} & , \text{ for model 2} \end{cases} . \quad (9)$$

Alternatively, we use a sequential procedure comparable to Kapetanios (2005). In a first step, we search for a single break which we select according to the maximum absolute t-value of the break dummy coefficient θ_1 for M1

¹So, the asymptotic results for the Perron-test in the case of a known break date apply also to the new test.

and κ_1 for M2 under the restriction of $\theta_2 = \delta_2 = 0$ for M1 and $\kappa_2 = \delta_2^* = \gamma_2^* = 0$ for M2:

$$\hat{T}_{B,1} = \begin{cases} \arg \max_{T_{B,1}} |t_{\hat{\theta}_1}(T_{B,1})| & , \text{ for model 1} \\ \arg \max_{T_{B,1}} |t_{\hat{\kappa}_1}(T_{B,1})| & , \text{ for model 2} \end{cases} . \quad (10)$$

So, in the first step, the test procedure reduces to the case described in Popp (2008). Under the restriction of the first break $\hat{T}_{B,1}$, we estimate the second break date $\hat{T}_{B,2}$ analogously to the first break. The results of the simultaneous and the sequential procedure do not differ much. So, we prefer the sequential procedure because it is far less computationally intensive. In the grid search case, we compute the test statistic approximately T^2 times compared to approximately $2T$ for the sequential procedure.

As discussed intensively by VP (1998) for the one break case, the Perron-type test statistics are invariant under the null hypothesis to a break in level and slope asymptotically as well as in finite samples when the break point is known. Because, as mentioned above, the procedure proposed by VP (1998) generalized to the two break case and the new procedure are identical for known break dates, the invariance results apply to the new unit root test. However, when the break dates are unknown, the invariance to level shifts for the Perron-type test no longer holds in finite samples leading

to considerable spurious rejections of the unit root null hypothesis, see VP (1998) and LS (2001). Moreover, the Perron-type test statistic capable of trend breaks is no longer invariant to breaks in slope neither in finite samples nor asymptotically, see VP (1998). In contrast, the invariance to level and slope breaks holds for the minimum LM unit root test proposed by LS (2003).

Because the spurious rejection property of existent ADF-type tests is primarily a problem in finite samples and for this reason a major drawback of their applicability, one main goal of the present paper is to show that the new ADF-type test are (approximately) invariant to level and slope breaks in finite samples by means of Monte Carlo simulations whose results are summarized in the following section.

3 Monte Carlo simulation results

All simulations were carried out in GAUSS 8.0. The series y_t is generated according to (1)-(3) together with (4) for M1 and (5) for M2 assuming the innovation process e_t to be standard normally distributed, $e_t \sim n.i.d.(0, 1)$. For e_t , samples of size $T + 50$ are generated, of which the first 50 observations are then discarded. Because our main focus is on the effect of varying break magnitudes on the test performance, we adopt the assumption made in comparable studies by VP (1998), Harvey et al. (2001) and LS (2001) and

set $\Psi^*(L) = 1$. The tests are conducted using (7) and (8) always assuming the appropriate lag order of $k = 0$ to be known.

3.1 Critical values

The critical values (CVs) are based on 50000 replications. For the M1- and M2-type tests, we calculate the CVs at the 1 per cent, 5 per cent, and 10 per cent levels for both the case of known and unknown break dates which we denote CVexo and CVendo, respectively. We generate CVs for sample sizes of $T = 50, 100, 300$, and 500. All CVs are calculated assuming no break, i.e. $\theta_i = 0$ in (4) for M1 and $\theta_i = \gamma_i = 0$ in (5) for M2, $i = 1, 2$. For the case of known break dates, we generate the dummy variables in (7) and (8) according to $T'_{B,i} = [\lambda'_i T]$, $i = 1, 2$, $[\cdot]$: greatest integer function, with the break fraction $\lambda' = (\lambda'_1, \lambda'_2) = (0.2, 0.4), (0.2, 0.6), (0.2, 0.8), (0.4, 0.6), (0.4, 0.8)$, and $(0.6, 0.8)$. For the case of unknown break points, we determine the break dates assuming that there exist two periods for M1 and three periods for M2 between the first and second break. The CVs for the case of known break dates are reported in Tables 1 for M1 and 2 for M2 and in the case of unknown break dates in Table 3.

It can be observed that CVexo vary only slightly with the break fraction λ' and that the CVexos for different break fractions converge as T increases

from $T = 50$ to $T = 500$. Furthermore, it can be seen that CVexo converges sharply to CVendo for the respective model with the sample size. This feature can be motivated in the following way. If the unit root test for unknown break dates is invariant to the break magnitude and the probability of detecting the true break dates goes to 1 with increasing break magnitude, i.e. for sufficiently large breaks we always identify the break dates correctly which corresponds to the situation of knowing the break dates, the distribution of the test statistic for known break dates has to coincide with the distribution of the test statistic for unknown break dates and consequently CVendo is equal to CVexo.

Both the break dates estimation accuracy and the invariance to level and slope breaks will be shown in the next subsection for the new unit root test.

3.2 Finite sample size

Because of the great computational burden, the simulations of the empirical size and power are based only on 5000 replications. For the size and power simulations, ρ is set to 1 and 0.9, respectively. The results for the size effects are reported in Tables 4 and 5 for models M1 and M2, respectively. We calculate the empirical size and power for the case of $\lambda' = (0.4, 0.6)$ and sample sizes of $T = 50, 100, 300$, and 500. We also generate results for

various combinations of the break fractions λ'_1 and λ'_2 using CVendo in Table 3 which turn out to be qualitatively equal.² This is evidence that the unit root test for unknown break points do not depend considerably on the break fraction parameters in finite samples.

We calculate the empirical size and power of the new unit root test for the case where the true break date is exogenously given (denoted 'exo' in Table 4 and 5) and for unknown break dates where we detect the break dates endogenously (denoted 'endo'). Furthermore, because of the relationship between CVexo and CVendo, we use CVexo for test decision in the unknown break dates case (denoted 'endoCVexo'). Thereby, we are able to show the correspondence of CVexo and CVendo.

The performance of the new test for M1 and M2 are similar. In the case of the exogenous break test the empirical size is independent of the magnitude of the breaks close to the nominal 5 per cent level proving the invariance to level and slope break for known break dates. The empirical size of the endogenous break test is also close to the nominal 5 per cent level in the case of a small break, but as the break magnitude increases the empirical size decreases slightly. The endogenous break test using CVexo, however, is a little bit oversized for small breaks and small sample sizes, but the size

²Due to space considerations, we only report results for the case $\lambda' = (0.4, 0.6)$; the rest of the results are available from the authors upon request.

converges to the 5 per cent nominal level with increasing break and sample size. The ability of the test to identify both breaks simultaneously is high even for medium sized breaks. Because we assume the realistic case of a fixed break size (independent of the sample size T), the probability decreases with the sample size as can be expected.

3.3 Empirical power

The empirical power of M1 and M2 are reported in the second half of Table 4 and in Table 6, respectively. The power of the exogenous break test and the endogenous break test do not differ substantially. This means that the additional information about the timing of the break do not augment the power of the test considerably. This is in contrast to the statement of Perron (1997) that a procedure imposing no a priori information with respect to the choice of the break date has relatively low power.

Moreover, the power of the test converges to 100 per cent with increasing sample size showing the consistency of the test. The results also reveal that the probability of detecting the true break date goes rapidly to 100 per cent with increasing break magnitude.

4 Application

In this section, we demonstrate the applicability of our proposed new models M1 and M2. We use two datasets on the US macroeconomic variables. The first dataset is the famous and commonly used Nelson and Plosser dataset. The second dataset is one that we compile from the *International Financial Statistics*, published by the International Monetary Fund.

There are two main differences between the Nelson and Plosser dataset and our new dataset. First, the Nelson and Plosser dataset considers data that includes the World War period, while our new dataset considers data in the post-war period. Our dataset is also the most up-to-date: the Nelson-Plosser dataset ends in 1970 while our dataset ends in either 2006 or in most cases 2007. It follows that the new dataset captures the most recent (over the last three to four decades) developments in the US economy, which may have implications for unit root testing. In any case, our aim here is not to draw on the economic theory that motivates a test for a unit root, rather it is to merely demonstrate the applicability of our test. The second difference is that Nelson and Plosser consider only 14 macroeconomic series, while the new dataset allows us to test for unit roots in 32 macroeconomic variables.

We begin with a discussion of results obtained from the Nelson and Plosser dataset. The results are reported in Table 7. Results from M1 reveal that

we are able to reject the unit root null hypothesis for GNP at the 1 per cent level, and for industrial production and the unemployment rate at the 10 per cent level. Finally, results from M2 reveal that we are able to reject the unit root null hypothesis for real GNP, industrial production, and real wage rates, all at the 5 per cent level. Taken together, results from our two models are able to reject the unit root null hypothesis for six out of 14 series, representing about 43 per cent of the variables considered.

In Table 8, we report results from our new dataset. All data series are converted into logarithmic form before the empirical analysis. The presentation of results is as follows. Column 1 lists the data series, column 2 contains results for M1, while column 3 contains results from the M2 model. For each of these respective models, test statistics for the null of a unit root, structural breaks, and optimal lag lengths are presented. The optimal lag length k is obtained by using the procedure suggested by Hall (1994).

Beginning with the M1 model, we find that we are able to reject the unit root null hypothesis for the unemployment rate, exports, the mortgage rate, and the export price index at the 10 per cent level, and for the T-bill rate at the 1 per cent level.

In the case of M2, we find that we are able to reject the unit root null hypothesis for M2, industrial production, the PPI (for capital equipment) and consumer goods at the 10 per cent level, for the unemployment rate,

mortgage rate and the NASDAQ index at the 5 per cent level, and for the CPI and the T-Bill rate at the 1 per cent level.

In sum, we find that based on models M1 and M2, we are able to reject the unit root null hypothesis for 13 of the 32 series. This represents about 41 per cent of the US macroeconomic series considered here. It is worth highlighting here that it is up to the applied researcher to choose the best model, which, in our view, should be dictated by economic theory.

5 Concluding remarks

In this paper, we proposed a new test for unit roots that is flexible enough to allow for at most two structural breaks in the level and trend of a data series. More specifically, we considered two different models for trending data: model 1 allows for two breaks in the level of the series and model 2 accounts for two breaks in the level and slope.

The key features of our test are that it is a ADF-type innovational outlier unit root test for which we specify the data generating process as an unobserved components model, and breaks are allowed under both the null and alternative hypotheses. Using Monte Carlo simulations, we showed that our proposed test has correct size, stable power, and identifies the structural breaks accurately.

We demonstrated the applicability of our unit root test through undertaking two exercises: one based on the Nelson and Plosser dataset and the other based on an updated post-war dataset. Using the new dataset, we found that tests based on models 1 and 2 taken together were able to reject the unit root null hypothesis for 13 of the 32 US macroeconomic series.

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Table 1: 1%, 5% and 10% critical values for exogenous two break test, Model 1, 50000 replications

T	λ_1	$\lambda_2 = 0.4$			$\lambda_2 = 0.6$			$\lambda_2 = 0.8$		
		1%	5%	10%	1%	5%	10%	1%	5%	10%
50	0.2	-4.953	-4.194	-3.826	-4.842	-4.127	-3.777	-4.895	-4.178	-3.827
	0.4	-	-	-	-4.850	-4.148	-3.780	-4.872	-4.145	-3.788
	0.6	-	-	-	-	-	-	-4.922	-4.191	-3.823
100	0.2	-4.760	-4.113	-3.787	-4.738	-4.077	-3.733	-4.761	-4.112	-3.785
	0.4	-	-	-	-4.745	-4.078	-3.741	-4.715	-4.087	-3.743
	0.6	-	-	-	-	-	-	-4.736	-4.112	-3.785
300	0.2	-4.664	-4.073	-3.770	-4.615	-4.037	-3.727	-4.642	-4.051	-3.754
	0.4	-	-	-	-4.620	-4.036	-3.721	-4.621	-4.039	-3.724
	0.6	-	-	-	-	-	-	-4.650	-4.067	-3.754
500	0.2	-4.640	-4.069	-3.759	-4.612	-4.024	-3.728	-4.624	-4.064	-3.755
	0.4	-	-	-	-4.600	-4.024	-3.713	-4.603	-4.023	-3.717
	0.6	-	-	-	-	-	-	-4.611	-4.058	-3.755

Table 2: 1%, 5% and 10% critical values for exogenous two break test, Model 2, 50000 replications

T	λ_1	$\lambda_2 = 0.4$			$\lambda_2 = 0.6$			$\lambda_2 = 0.8$		
		1%	5%	10%	1%	5%	10%	1%	5%	10%
50	0.2	-5.401	-4.609	-4.221	-5.635	-4.866	-4.501	-5.390	-4.616	-4.231
	0.4	-	-	-	-5.591	-4.876	-4.499	-5.645	-4.882	-4.507
	0.6	-	-	-	-	-	-	-5.380	-4.631	-4.251
100	0.2	-5.232	-4.577	-4.237	-5.404	-4.768	-4.450	-5.252	-4.602	-4.252
	0.4	-	-	-	-5.430	-4.782	-4.457	-5.387	-4.784	-4.462
	0.6	-	-	-	-	-	-	-5.246	-4.574	-4.231
300	0.2	-5.135	-4.537	-4.224	-5.276	-4.720	-4.421	-5.163	-4.557	-4.236
	0.4	-	-	-	-5.279	-4.724	-4.420	-5.297	-4.712	-4.418
	0.6	-	-	-	-	-	-	-5.140	-4.549	-4.238
500	0.2	-5.125	-4.541	-4.233	-5.251	-4.699	-4.410	-5.126	-4.544	-4.239
	0.4	-	-	-	-5.273	-4.712	-4.415	-5.271	-4.712	-4.409
	0.6	-	-	-	-	-	-	-5.136	-4.534	-4.219

Table 3: 1%, 5% and 10% critical values for endogenous two break test (computed under the assumption of no breaks), 50000 replications

T	M1			M2		
	1%	5%	10%	1%	5%	10%
50	-5.259	-4.514	-4.143	-5.949	-5.181	-4.789
100	-4.958	-4.316	-3.980	-5.576	-4.937	-4.596
300	-4.731	-4.136	-3.825	-5.318	-4.741	-4.430
500	-4.672	-4.081	-3.772	-5.287	-4.692	-4.396

Table 4: 5 percent rejection frequency with nominal 5 percent significance level and probability of detecting the true break date, ML, $\lambda' = (0.4, 0.6)$, 5000 replications

T	θ	empirical size ($\rho = 1$)				empirical power ($\rho = 0.9$)			
		exo	endo	endoCVexo	$P(T_B = T'_B)$	exo	endo	endoCVexo	$P(T_B = T'_B)$
50	0	0.050	0.050	0.089	0.003	0.067	0.068	0.128	0.003
50	3	0.045	0.042	0.073	0.498	0.066	0.061	0.102	0.474
50	5	0.047	0.027	0.051	0.970	0.068	0.043	0.077	0.955
50	10	0.045	0.024	0.045	1.000	0.059	0.028	0.059	1.000
50	20	0.046	0.023	0.046	1.000	0.066	0.031	0.066	1.000
100	0	0.050	0.050	0.083	0.000	0.147	0.136	0.199	0.000
100	3	0.047	0.038	0.063	0.411	0.133	0.102	0.155	0.397
100	5	0.055	0.034	0.057	0.969	0.139	0.087	0.140	0.960
100	10	0.050	0.030	0.050	1.000	0.139	0.083	0.139	1.000
100	20	0.055	0.031	0.055	1.000	0.133	0.083	0.133	1.000
300	0	0.050	0.050	0.061	0.000	0.789	0.762	0.801	0.000
300	3	0.048	0.044	0.055	0.278	0.789	0.690	0.737	0.284
300	5	0.048	0.039	0.049	0.943	0.791	0.733	0.781	0.938
300	10	0.046	0.036	0.046	1.000	0.775	0.731	0.775	1.000
300	20	0.045	0.037	0.045	1.000	0.780	0.733	0.780	1.000
500	0	0.050	0.050	0.052	0.000	0.998	0.997	0.998	0.000
500	3	0.052	0.051	0.056	0.236	0.999	0.992	0.993	0.218
500	5	0.046	0.044	0.047	0.921	0.997	0.994	0.994	0.919
500	10	0.048	0.045	0.048	1.000	0.998	0.998	0.998	1.000
500	20	0.050	0.047	0.050	1.000	0.999	0.998	0.999	1.000

Table 5: 5 percent rejection frequency with nominal 5 percent significance level and probability of detecting the true break date, M2, $\lambda' = (0.4, 0.6)$, 5000 replications

T	θ	γ	empirical size ($\rho = 1$)			$P(\hat{T}_B = T'_B)$
			exo	endo	endoCVexo	
50	0	0	0.050	0.050	0.092	0.003
50	0	5	0.048	0.033	0.053	0.455
50	0	10	0.053	0.026	0.049	0.893
50	5	0	0.052	0.026	0.057	0.948
50	5	5	0.050	0.026	0.050	1.000
50	5	10	0.053	0.026	0.053	1.000
50	10	0	0.051	0.023	0.051	1.000
50	10	5	0.054	0.028	0.054	1.000
50	10	10	0.053	0.025	0.053	1.000
100	0	0	0.050	0.050	0.069	0.001
100	0	5	0.051	0.068	0.079	0.292
100	0	10	0.049	0.029	0.040	0.763
100	5	0	0.051	0.036	0.051	0.955
100	5	5	0.047	0.033	0.047	1.000
100	5	10	0.049	0.037	0.048	1.000
100	10	0	0.049	0.037	0.049	1.000
100	10	5	0.053	0.038	0.053	1.000
100	10	10	0.050	0.038	0.050	1.000
300	0	0	0.050	0.050	0.056	0.000
300	0	5	0.059	0.091	0.093	0.097
300	0	10	0.055	0.031	0.035	0.405
300	5	0	0.057	0.052	0.058	0.940
300	5	5	0.055	0.050	0.055	1.000
300	5	10	0.056	0.050	0.056	1.000
300	10	0	0.052	0.044	0.052	1.000
300	10	5	0.051	0.045	0.051	1.000
300	10	10	0.058	0.050	0.058	1.000
500	0	0	0.050	0.050	0.057	0.000
500	0	5	0.049	0.069	0.071	0.057
500	0	10	0.051	0.017	0.021	0.250
500	5	0	0.055	0.045	0.055	0.920
500	5	5	0.048	0.040	0.048	0.999
500	5	10	0.056	0.049	0.056	1.000
500	10	0	0.054	0.046	0.054	1.000
500	10	5	0.053	0.047	0.053	1.000
500	10	10	0.053	0.045	0.053	1.000

Table 6: Empirical power of the M2 model

empirical power ($\rho = 0.9$)						
T	θ	γ	exo	endo	endoCVexo	$P(\bar{T}_B = T'_B)$
50	0	0	0.060	0.055	0.104	0.003
50	0	5	0.063	0.042	0.065	0.414
50	0	10	0.067	0.029	0.059	0.871
50	5	0	0.064	0.041	0.074	0.931
50	5	5	0.063	0.032	0.063	1.000
50	5	10	0.065	0.030	0.065	1.000
50	10	0	0.070	0.031	0.070	1.000
50	10	5	0.057	0.026	0.057	1.000
50	10	10	0.066	0.035	0.066	1.000
100	0	0	0.089	0.105	0.136	0.002
100	0	5	0.087	0.087	0.100	0.246
100	0	10	0.090	0.050	0.069	0.729
100	5	0	0.095	0.072	0.098	0.942
100	5	5	0.085	0.062	0.084	1.000
100	5	10	0.100	0.072	0.100	1.000
100	10	0	0.093	0.068	0.093	1.000
100	10	5	0.090	0.063	0.090	1.000
100	10	10	0.095	0.070	0.095	1.000
300	0	0	0.587	0.570	0.597	0.000
300	0	5	0.594	0.229	0.240	0.075
300	0	10	0.592	0.287	0.304	0.331
300	5	0	0.582	0.545	0.574	0.932
300	5	5	0.595	0.565	0.594	0.998
300	5	10	0.590	0.560	0.590	1.000
300	10	0	0.591	0.561	0.591	1.000
300	10	5	0.586	0.559	0.586	1.000
300	10	10	0.598	0.567	0.598	1.000
500	0	0	0.974	0.958	0.970	0.000
500	0	5	0.974	0.339	0.352	0.036
500	0	10	0.976	0.459	0.471	0.197
500	5	0	0.968	0.952	0.959	0.911
500	5	5	0.974	0.966	0.974	0.997
500	5	10	0.974	0.964	0.974	0.999
500	10	0	0.974	0.969	0.974	1.000
500	10	5	0.972	0.963	0.972	1.000
500	10	10	0.971	0.964	0.971	1.000

Table 7: Results of two-break unit root test, Nelson-Plosser data

Nr.	Series	Sample	T	M1			M2		
				test statistic	TB1	TB2	test statistic	TB1	TB2
1	Real GDP	1909-1970	62	-3.680	1929	1931	-5.597**	1921	1938
2	Nominal GNP	1909-1970	62	-6.396***	1929	1941	-3.705	1921	1940
3	Real per Capita GNP	1909-1970	62	-3.491	1929	1931	-5.529**	1921	1938
4	Industrial Production	1860-1970	111	-4.310*	1920	1931	-4.632	1920	1931
5	Employment	1890-1970	81	-2.002	1931	1945	-2.145	1931	1945
6	Unemployment	1890-1970	81	-4.130*	1917	1922	-3.703	1917	1923
7	GNP Deflator	1889-1970	82	-2.777	1916	1920	-2.749	1916	1920
8	Consumer Prices	1860-1970	111	-1.582	1916	1920	-2.733	1916	1920
9	Wages	1900-1970	71	-1.636	1920	1931	-3.160	1920	1940
10	Real Wages	1900-1970	71	-1.622	1931	1945	-5.565**	1931	1940
11	Money Stock	1889-1970	82	-2.029	1920	1931	-3.191	1920	1931
12	Velocity	1869-1970	102	-2.886	1941	1945	-4.228	1917	1941
13	Bond Yield	1900-1970	71	0.026	1921	1932	-0.247	1917	1931
14	Common Stock Prices	1871-1970	100	-1.928	1931	1937	-4.215	1931	1942

Table 8: Results of two-break unit root test using logarithmized data

Nr.	Series	Sample	T	M1				M2			
				test statistic		TB1	TB2	test statistic		TB1	TB2
					k				k		
1	Reer based on rel. CP	1980-2006	27	-1.713	0	1968	1974	-2.262	0	1968	1973
2	Reer based on RNULC	1975-2007	33	-2.787	4	1985	1999	-3.223	4	1985	1999
3	M1	1959-2007	49	-1.167	3	1985	1994	-3.113	3	1985	1994
4	M2	1959-2007	49	-2.643	1	1970	1974	-4.848*	1	1970	1986
5	M3	1959-2005	47	2.357	0	1968	1994	1.778	0	1969	1994
6	Gross saving	1948-2006	59	-2.593	0	1972	1983	-4.547	5	1977	1983
7	Gross national income	1948-2006	59	0.308	4	1981	1990	-2.958	4	1972	1981
8	Gross domestic product	1948-2007	60	0.067	5	1961	1981	-3.429	4	1972	1981
9	GDP Deflator	1948-2007	60	-2.504	4	1973	1976	0.899	2	1975	1981
10	Wages	1948-2007	60	4.678	0	1967	1981	-3.130	0	1967	1981
11	Industrial production	1950-2007	58	-1.360	5	1974	1979	-5.036*	4	1974	1983
12	Crude petroleum production	1948-2006	59	-2.221	0	1965	1977	-4.240	4	1973	1988
13	Non-agricultural employment	1948-2007	60	-2.587	0	1965	1981	-3.464	1	1974	1981
14	Unemployment rate	1948-2007	60	-4.337*	4	1974	1981	-5.374**	5	1974	1983
15	Exports	1948-2007	60	-4.331*	1	1972	1978	-4.168	1	1972	1981
16	Export price index	1948-2007	60	-4.157*	2	1972	1974	-4.485	2	1972	1975
17	Imports	1948-2007	60	-1.944	0	1973	1975	-0.775	2	1973	1981
18	Import price index	1948-2007	60	-1.817	3	1973	1978	0.1891	1	1973	1980
19	3-months T-Bill rate	1975-2007	33	-6.803***	5	1991	1999	-9.412***	1	1996	1999
20	Bank prime loan rate	1948-2007	60	-1.745	4	1974	1984	-7.436***	1	1980	1984
21	Mortgage rate	1972-2007	36	-4.158*	4	1979	1985	-5.686**	4	1982	1998
22	Bond yield, 3 year	1948-2007	60	-1.274	3	1980	1985	-4.691	2	1985	1993
23	Share prices index	1948-2007	60	-2.670	0	1982	1995	-2.989	0	1982	1985
24	NASDAQ composite index	1972-2007	36	-5.681***	0	1987	1999	-5.459**	4	1989	1998
25	AMEX average index	1971-2007	37	1.039	5	1981	1983	-0.967	5	1981	1984
26	S&P industrials index	1948-2007	60	-2.175	0	1969	1973	-3.565	4	1973	1982
27	PPI / WPI	1948-2007	60	-3.897	3	1972	1978	-4.849*	1	1972	1976
28	PPI: Capital equipment	1948-2007	60	-3.642	2	1973	1976	-4.096	2	1973	1976
29	WPI: finished goods	1948-2007	60	-3.118	3	1973	1978	-1.483	1	1973	1981
30	CPI: all items	1948-2007	60	-2.335	5	1972	1974	-6.745***	1	1968	1974
31	CPI: finished goods	1948-2007	60	-4.394*	1	1972	1978	-5.164*	1	1972	1976
32	Industrial goods index	1948-2007	60	-3.652	5	1973	1986	-2.214	1	1973	1981